Statistics

Chapter 14 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 15

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PROBLEM STATEMENT

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Solution A

Step 1

A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

The independent variable (x-variable) is always scaled on the horizontal axis, while the dependent variable (y-variable) is always scaled on the $vertical$ axis.

Result

Vertical axis

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Solution A

Step 1

Level of measurement

The level of measurement is **nominal**, if the values of the variables can only be named and not ordered.

The level of measurement is **ordinal**, if the values of the variables can be named and ordered, but the difference between two values has no meaning.

The level of measurement is **interval**, if the difference between two values has a meaning, but the variable contains no absolute zero.

The level of measurement is **ratio**, if the variable contains an absolute zero.

The correlation coefficient can only be calculated when the data values are numerical (with numerical meaning). This then also implies that the difference between two data values needs to have meaning as well (as we use differences in the calculation of the correlation coefficient) and thus we require at least the *interval* level of measurement for the data (that is, the level of measurement of both variables is either interval and/or ratio).

Result

Interval

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Solution A

Step 1

The correlation coefficient measures the strength of the linear relationship between two variables.

When there is no correlation between two variables, then this implies that there is no linear relationship between the variables and thus the correlation coefficient is approximately **zero**.

Result

0

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Solution A

Step 1

The correlation coefficient measures the strength of the linear relationship between two variables.

Moreover, the correlation coefficient can take on values from -1 to $+1$.

The strongest correlation between two variables is then the correlation that is the largest in absolute value (that is, the value that is closest to -1 or $+1$).

In this case, we note that the correlation coefficient -0.77 is the largest in absolute value and thus the value -0.77 indicates the strongest correlation between two variables.

Result

$$-0.77$$

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Solution A

Step 1

The correlation coefficient r can take on values from -1 to $+1$.

Since the coefficient of determination r^2 is the square of the correlation coefficient r, the coefficient of determination can take on values from 0 to 1.

This then implies that it is $\mathbf{impossible}$ that the coefficient of determination is greater than 1.

Result

Impossible

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The point at which the regression equation crosses the y-axis always has x-coordinate 0 (as the y-axis is the line $x = 0$).

The y-coordinate of this point is the y-intercept, which is the constant in the regression equation and thus the y-coordinate of this point is 7.

This then implies that the point where the regression equation crosses the y-axis is $(0, 7)$.

Result

$$(0, 7)$$

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Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The slope of the regression equation is the coefficient of x:

$$b = -0.5$$

The slope of the regression equation represents the average increase in the dependent y-variable per unit of the independent x-variable.

Thus we expect the dependent variable to decrease by 0.5 units when the independent variable increases by 1 unit.

Result

Decrease by 0.5 units

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The slope of the regression equation is the coefficient of x:

$$b = -0.5$$

Since the slope $b = -0.5$ is negative, there is a negative relationship between the variables and thus the correlation coefficient needs to be negative as well.

The correlation coefficient r is then the negative square root of the coefficient of determination.

$$r = -\sqrt{r^2} = -\sqrt{0.81} = -0.9$$

Result

$$r = -0.9$$

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Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The standard error of the estimate measures the average deviation between the data points and the regression line.

When all data points are on the regression line, then there are no deviations between the data points and the regression line, which implies that the standard error of the estimate then needs to be 0.

Result

0

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PROBLEM STATEMENT

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Solution A

Step 1

There is no limitation on the number of independent variables that you are allowed to use in a multiple regression equation and thus the maximum number of independent variables allowed is **unlimited**.

Result

Unlimited

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PROBLEM STATEMENT

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Solution A

Step 1

We always assume that there exists a **linear** relationship between the dependent variable and the set of independent variables within a multiple regression analysis.

Result

Linear

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Solution A

Step 1

A **residual** is the difference between the observed Y-value and the predicted Y-value $\hat{Y}$.

Result

Residual

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PROBLEM STATEMENT

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Solution A

Step 1

A dummy variable always has **two** possible outcomes: 0 and 1.

1 represents the occurrence of some event, while 0 represents that the event does not occur.

Result

Two

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PROBLEM STATEMENT

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Solution A

Step 1

A **correlation matrix** is a table that shows all possible correlation coefficients between the dependent variable and each independent variable, and between each pair of independent variables.

Result

Correlation matrix

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Solution A

Step 1

A graph of the residuals will have approximately a **normal distribution** when there exists a linear relationship between the dependent variable and the set of independent variables.

Result

Normal distribution